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Title

Implementation of Vacuum Cleaner Agent using Artificial Intelligence

Aim:

To design and implement an intelligent Vacuum Cleaner Agent that perceives the condition of rooms in a school environment and takes appropriate actions to clean dirty or wet rooms while maximizing its performance score.

Objectives:

To understand the concept of an Intelligent Agent. To implement an agent interacting with an environment. To identify different room conditions such as Clean, Dirty, and Wet. To make the agent take appropriate cleaning actions. To calculate the agent's performance using rewards and movement costs. To understand the PEAS framework of an intelligent agent.

Outcomes

After completing this experiment, the following outcomes were achieved:

Successfully implemented a Vacuum Cleaner Intelligent Agent using Python. Understood the interaction between an agent and its environment. Implemented a school environment containing multiple floors and rooms. The agent successfully identifies three room conditions: Clean, Dirty, and Wet. The agent takes appropriate actions based on the room condition: Dirty → Clean Wet → Dry and Clean Clean → No Action Implemented a performance measure using rewards and movement costs. Understood and applied the PEAS framework to an intelligent agent. Verified that after the agent completes its task, all rooms are converted to the Clean state. Learned how an intelligent agent uses perception, decision-making, and action to

achieve its goal. Demonstrated a basic goal-based/rational agent operating in a dynamic environment.

Conceptual Details

Intelligent Agent

An intelligent agent is a system that perceives its environment through sensors and performs actions through actuators to achieve a particular goal.

In this experiment:

Agent: Vacuum Cleaner Agent

Environment: School

The agent continuously observes the state of the rooms and performs actions based on the observed condition.

Environment States

Each room can have one of three states:

1.Clean

2.Dirty

3.Wet

4.Agent Actions

The agent can perform the following actions:

1.Move to Floor,

2.Move to Room,

3.Clean Dirty Room,

4.Dry/Clean Wet Room,

5.Do Nothing if Room is Clean.

✓ Algorithm

Algorithm: Vacuum Cleaner Agent

Step 1: Start the program.

Step 2: Create the school environment with two floors.

Step 3: Create four rooms on each floor.

Step 4: Randomly assign each room one of the following states:

Clean / Dirty / Wet

Step 5: Display the initial status of all rooms.

Step 6: Initialize the vacuum cleaner agent and set its score to zero.

Step 7: Move the agent to the first floor.

Step 8: Check each room on the floor.

Step 9: If the room is Dirty:

Clean the room. Change its state to Clean. Add +10 to the score.

Step 10: If the room is Wet:

Dry and clean the room. Change its state to Clean. Add +10 to the score.

Step 11: If the room is already Clean:

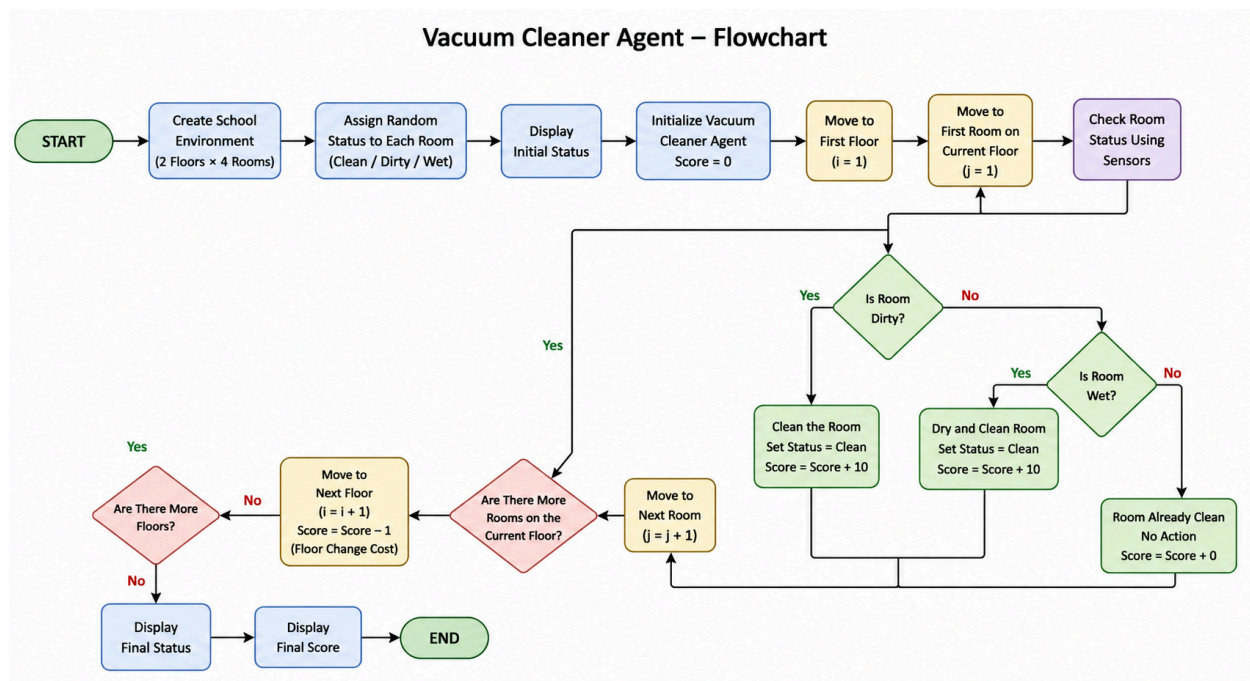
Take no cleaning action. Continue to the next room.

Step 12: Move to the next floor and repeat the process.

Step 13: After checking all rooms, display the final performance score.

Step 14: Display the final status of the school.

Step 15: Stop.



```
import random
```

```
# Environment
class School:
```

```

def __init__(self):
    self.floors = {
        "Floor 1": {
            "Room 1": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 2": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 3": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 4": random.choice(["Dirty", "Clean", "Wet"]),
        },
        "Floor 2": {
            "Room 1": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 2": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 3": random.choice(["Dirty", "Clean", "Wet"]),
            "Room 4": random.choice(["Dirty", "Clean", "Wet"]),
        }
    }

def display_status(self):
    print("\nSchool Status:")

    for floor, rooms in self.floors.items():
        print(f"\n{floor}")

        for room, status in rooms.items():
            print(f"    {room}: {status}")

# Agent
class VacuumCleanerAgent:
    def __init__(self, school):
        self.school = school
        self.score = 0

    def clean_school(self):
        print("\nVacuum Cleaner Agent Started...\n")

        for floor, rooms in self.school.floors.items():

            print(f"\nMoving to {floor}")
            self.score -= 1 # Cost for changing floor

            for room, status in rooms.items():

                print(f"Checking {room}... ", end="")
                self.score -= 1 # Cost for moving to a room

                if status == "Dirty":
                    print("Dirty -> Cleaning")
                    self.school.floors[floor][room] = "Clean"
                    self.score += 10 # Reward for cleaning

                elif status == "Wet":
                    print("Wet -> Drying and Cleaning")
                    self.school.floors[floor][room] = "Clean"
                    self.score += 10 # Reward for cleaning

```

```
        else:
            print("Already Clean")

    print("\nCleaning Completed!")
    print("Final Performance Score:", self.score)

# Main Program
school = School()

print("Initial Status:")
school.display_status()

agent = VacuumCleanerAgent(school)
agent.clean_school()

print("\nFinal Status:")
school.display_status()
```

Initial Status:

School Status:

Floor 1

Room 1: Dirty
Room 2: Wet
Room 3: Dirty
Room 4: Dirty

Floor 2

Room 1: Wet
Room 2: Dirty
Room 3: Dirty
Room 4: Dirty

Vacuum Cleaner Agent Started...

Moving to Floor 1

Checking Room 1... Dirty -> Cleaning
Checking Room 2... Wet -> Drying and Cleaning
Checking Room 3... Dirty -> Cleaning
Checking Room 4... Dirty -> Cleaning

Moving to Floor 2

Checking Room 1... Wet -> Drying and Cleaning
Checking Room 2... Dirty -> Cleaning
Checking Room 3... Dirty -> Cleaning
Checking Room 4... Dirty -> Cleaning

Cleaning Completed!

Final Performance Score: 70

Final Status:

School Status:

Floor 1

```
Room 1: Clean
Room 2: Clean
Room 3: Clean
Room 4: Clean

Floor 2
Room 1: Clean
Room 2: Clean
Room 3: Clean
Room 4: Clean
```

Result and Analysis

The Vacuum Cleaner Agent successfully operates in the school environment and identifies the condition of each room. The agent takes appropriate actions based on the condition of the room.

If a room is **Dirty**, the agent cleans it and changes its status to **Clean**. If a room is **Wet**, the agent dries and cleans it, changing its status to **Clean**. If a room is already **Clean**, the agent takes no action.

After completing the cleaning process, **all rooms in the school become Clean**. The agent also maintains a performance score based on the rewards received for cleaning and the costs incurred while moving between floors and rooms.

For example, if the agent cleans **5 rooms**, the cleaning reward is:

$$\text{Cleaning Reward} = 5 \times 10 = 50$$

If the agent incurs **10 movement/checking costs**, the movement cost is:

$$\text{Movement Cost} = 10 \times (-1) = -10$$

Therefore, the final performance score is:

$$\text{Final Score} = 50 - 10 = 40$$

The exact final score may vary in each execution because the initial condition of every room is generated randomly. The experiment demonstrates that the agent can effectively perceive the environment, make decisions, perform cleaning actions, and achieve its goal of maintaining a clean school environment.

Conclusion

The Vacuum Cleaner Agent was successfully implemented using Python. The agent interacts with a school environment, senses the condition of each room, and performs appropriate actions.

The experiment demonstrates the basic concepts of Artificial Intelligence, Intelligent Agents, Environment, Perception, Action, and Performance Measurement.

The addition of the Wet condition makes the agent more capable because it can handle both dirty and wet rooms instead of only detecting whether a room is clean or dirty.

Hence, the experiment successfully demonstrates the working of an intelligent vacuum cleaner agent using the PEAS framework.

Github Link

https://github.com/Samu7709/AIML_Lab_Assignments_CA